

Notes: Brunnermeier, Merkel & Sannikov (JPE, 2024)

Safe Assets

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1 Big picture

- **Question.** What makes government debt a safe asset, why can it have a negative beta, and how much government debt can the economy absorb?
- **Main idea.** A safe asset is valuable not only because of its cash flows but also because it provides a *service flow*: households can trade it after idiosyncratic shocks to partially self-insure.
- **Good-friend analogy.** A safe asset is valuable because it is “around” exactly when agents need it. After a bad private shock, an agent can sell or hold the safe asset to smooth consumption.
- **Incomplete markets.** Households operate firms subject to undiversifiable idiosyncratic risk. A skin-in-the-game constraint prevents complete risk sharing, so safe assets help households manage residual risk.
- **Dynamic trading perspective.** The value of government debt can be decomposed into discounted cash flows plus discounted service flows generated by equilibrium re trading.
- **Negative beta.** If idiosyncratic risk rises in recessions, the value of the safe asset’s service flow rises in recessions. This can make government debt appreciate in bad times and become a negative-beta asset.
- **Debt valuation puzzle.** Standard debt valuation using only future primary surpluses can make government debt look overvalued. The service-flow component can fill the gap.
- **Exorbitant privilege.** The safe-asset service flow lowers the government’s required cash-flow return, allowing government debt to be sustained even when primary surpluses are small or negative.
- **Debt Laffer curve.** A government can “mine” the safe-asset bubble by issuing debt faster, but more mining erodes the value of debt. The model therefore has a maximum sustainable permanent deficit.
- **Quantitative result.** In the calibrated dynamic model, the peak sustainable primary deficit is about 2.25% of GDP; without time-varying countercyclical idiosyncratic risk, the comparable peak is only about 0.1%.
- **Asset-pricing implication.** Flight to safety into government bonds can also help explain why outside equity is risky and volatile: insiders require higher compensation for bearing firm-specific risk in recessions.

2 Incremental contribution

- **New valuation object.** The paper introduces a dynamic-trading valuation formula for safe assets in incomplete markets. Standard valuation prices buy-and-hold cash flows; this paper prices the cash flows of equilibrium trading strategies.
- **Service flow without utility convenience.** Many safe-asset models put convenience yield directly into preferences or transaction technology. This paper derives the convenience-like value from the ability to retrade government debt for self-insurance.
- **Unifies three puzzles.** The model jointly addresses the debt valuation puzzle, the negative-beta property of safe assets, and the possibility of sustainable government debt with low or negative primary surpluses.
- **Bubbles and safety are complements.** A bubble can make an asset safer because a higher market value increases the service-flow component. This gives a precise meaning to the “safe asset tautology”: safe assets can be safe partly because they are expected to remain safe.
- **Public versus private safe assets.** Private assets can also generate service flows for their holders, but governments have a special ability to defend a debt bubble through taxation, regulation, and central-bank market-making.

- **Debt capacity as a Laffer curve.** The paper does not only ask whether debt can be bubbly. It quantifies how much permanent deficit can be financed by bubble mining and shows that the answer depends critically on countercyclical idiosyncratic risk.
- **Equity-market implication.** The same mechanism that makes government bonds safe makes outside equity risky: recessions increase idiosyncratic risk borne by insiders, lowering payouts to outside equityholders.

3 Data, calibration, and measurement

3.1 What is being calibrated

- The paper is a quantitative theory paper, not an empirical regression paper.
- The model is calibrated to US-style business-cycle and asset-pricing moments.
- Key targets include output volatility, consumption volatility, investment volatility, surplus volatility, the government debt-output ratio, the unlevered equity premium, and the equity Sharpe ratio.
- The state variable that matters most is time-varying idiosyncratic risk, denoted $\tilde{\sigma}_t$.

Interpretation: The quantitative exercise asks whether a model of self-insurance through government bonds can simultaneously resemble macro data, debt valuation, and asset-pricing facts.

3.2 Core objects

- K_t : aggregate capital.
- B_t : nominal government debt.
- P_t : price level.
- $q_t^B = (B_t/P_t)/K_t$: real value of government debt per unit of capital.
- q_t^K : market value of a unit of physical capital.
- ϑ_t : households' bond wealth share, approximately the relative price of bonds versus capital.
- $s_t = (\tau_t a_t - g)$: primary surplus per unit of aggregate capital.
- $\tilde{\sigma}_t$: idiosyncratic risk.
- $\bar{x}$: minimum undiversifiable exposure from the skin-in-the-game constraint.

Interpretation: The model's safe-asset mechanism lives in q_t^B and ϑ_t : as idiosyncratic risk rises, households demand more bond wealth, so the value of government debt rises relative to capital.

3.3 Exogenous risk and government policy

- Productivity a_t , idiosyncratic risk $\tilde{\sigma}_t$, debt policy $\bar{\mu}_t^B$, and the nominal rate i_t are modeled as aggregate Markov processes.
- Government spending is proportional to aggregate capital, gK_t .
- Taxes adjust to satisfy the government budget constraint.
- The quantitative Laffer curve varies the average interest-adjusted debt growth rate while keeping other parameters fixed.

Interpretation: The key comparative static is how much revenue the government can raise by expanding debt issuance when the private sector values debt as insurance.

4 Models and theoretical specifications

4.1 Baseline asset-pricing decomposition

The paper's headline pricing equation is

$$\text{price}_t = \mathbb{E}_t [\text{PV}_{r^{**}}(\text{cash flows})] + \mathbb{E}_t [\text{PV}_{r^{**}}(\text{service flows})]. \quad (1)$$

Interpretation: The first term is the standard payoff value. The second term is the insurance value from being able to retrade the asset after shocks. The discount rate r^{**} is the relevant rate from the dynamic-trading perspective.

Since now that results from the ability to self-insure through trading. The real value of a safe asset (or any tradable asset) is

$$\text{price}_t = \mathbb{E}_t [\text{PV}_{r^{**}}[\text{cash flows}]] + \mathbb{E}_t [\text{PV}_{r^{**}}[\text{service flows}]].$$

Original paper equation panel: cash-flow value plus service-flow value.

4.2 Preferences and production

Households have log preferences,

$$V_0^i = \mathbb{E} \left[\int_0^\infty e^{-\rho t} \log c_t^i dt \right]. \quad (2)$$

Firm i 's capital evolves as

$$\frac{dk_t^i}{k_t^i} = (\Phi(i_t^i) - \delta) dt + \tilde{\sigma}_t d\tilde{Z}_t^i + d\Delta_t^{k,i}. \quad (3)$$

Interpretation: Idiosyncratic shocks make private capital risky. Trading $d\Delta_t^{k,i}$ lets agents adjust positions, but skin-in-the-game prevents full diversification.

4.3 Government budget and resource constraint

The nominal government budget constraint is

$$i_t B_t + P_t g K_t = \mu_t^B B_t + P_t \tau_t a_t K_t. \quad (4)$$

The aggregate resource constraint is

$$C_t + g K_t + \iota_t K_t = a_t K_t. \quad (5)$$

Interpretation: Government debt is not simply an accounting liability. Its real value is endogenously determined by private demand for self-insurance and by expected fiscal policy.

4.4 Skin-in-the-game constraint

Agents can sell outside equity but must retain enough undiversified exposure:

$$-v_t^{E,i} \leq (1 - \bar{x}) v_t^{K,i}. \quad (6)$$

Interpretation: This is the friction that prevents full insurance. Without it, households could diversify idiosyncratic firm risk completely, and government bonds would lose the service-flow value emphasized in the paper.

4.5 Equilibrium prices as functions of bond share

In equilibrium, the investment rate, bond value, and capital value are

$$\iota_t = \frac{(1 - \vartheta_t)(a_t - g) - \rho}{1 - \vartheta_t + \phi\rho}, \quad (7)$$

$$q_t^B = \vartheta_t \frac{1 + \phi(a_t - g)}{1 - \vartheta_t + \phi\rho}, \quad (8)$$

$$q_t^K = (1 - \vartheta_t) \frac{1 + \phi(a_t - g)}{1 - \vartheta_t + \phi\rho}. \quad (9)$$

Interpretation: Once ϑ_t is known, the rest of the equilibrium follows. Higher ϑ_t means households want more of their wealth in bonds, increasing the relative value of government debt.

4.6 The bond-share equation

The equilibrium bond share solves the backward stochastic differential equation

$$\mathbb{E}_t[d\vartheta_t] = [\rho + \bar{\mu}_t^B - (1 - \vartheta_t)^2 \bar{x}^2 \tilde{\sigma}_t^2] \vartheta_t dt. \quad (10)$$

Interpretation: The final term is the insurance demand force. When idiosyncratic risk is high, safe assets are more valuable because households need a stronger hedge against undiversified shocks.

4.7 Buy-and-hold valuation

The standard buy-and-hold perspective gives

$$\frac{B_0}{P_0} = \lim_{T \rightarrow \infty} \mathbb{E} \left[\int_0^T \bar{\xi}_t s_t K_t dt \right] + \mathbb{E} \left[\bar{\xi}_T \frac{B_T}{P_T} \right]. \quad (11)$$

Interpretation: This formula prices future primary surpluses plus a possible terminal bubble. It is standard but can create the debt valuation puzzle if primary surpluses are procyclical and too small.

$\bar{\xi} = \mathbf{0}$ satisfies

$$\frac{B_0}{P_0} = \lim_{T \rightarrow \infty} \left(\mathbb{E} \left[\int_0^T \bar{\xi}_t s_t K_t dt \right] + \mathbb{E} \left[\bar{\xi}_T \frac{B_T}{P_T} \right] \right). \quad (11)$$

or consists of two terms: a discounted stream of primary

Original paper equation panel: buy-and-hold government debt valuation.

4.8 Dynamic trading valuation

The dynamic trading perspective gives

$$\frac{B_0}{P_0} = \mathbb{E} \left[\int_0^\infty \xi_t^{**} s_t K_t dt \right] + \mathbb{E} \left[\int_0^\infty \xi_t^{**} (1 - \vartheta_t)^2 \bar{x}^2 \tilde{\sigma}_t^2 \frac{B_t}{P_t} dt \right]. \quad (12)$$

Interpretation: The second term is the key service-flow term. It is proportional to the value of public debt and to the undiversifiable idiosyncratic risk that households must bear.

ment debt at $t = 0$ satisfies

$$\frac{\mathcal{B}_0}{\mathcal{P}_0} = \mathbb{E} \left[\int_0^\infty \underbrace{\left(\int \xi_t^i \eta_t^i di \right)}_{=\xi_t^{**}} s_t K_t dt \right] + \mathbb{E} \left[\int_0^\infty \underbrace{\left(\int \xi_t^i \eta_t^i di \right)}_{=\xi_t^{**}} (1 - \vartheta_t)^2 \bar{\chi}^2 \bar{\sigma}_t^2 \frac{\mathcal{B}_t}{\mathcal{P}_t} dt \right]. \quad (13)$$

Original paper equation panel: dynamic-trading government debt valuation.

4.9 Debt Laffer curve

Primary deficits per unit of capital satisfy the simple relationship

$$-s_t = \bar{\mu}_t^B q_t^B. \quad (13)$$

Interpretation: The debt growth rate is like a tax rate on bondholders, and the value of debt is the tax base. Raising debt growth can increase revenue, but it also reduces the value of debt. This creates a Laffer curve.

5 Conceptual design and credibility

5.1 What the model explains

- Why government bonds can have low cash-flow returns.
- Why government bonds can be negative-beta assets.
- Why the value of government debt can exceed the value implied by discounted primary surpluses.
- Why safe-asset status can be self-reinforcing and bubbly.
- Why there is still a limit to debt-financed deficits.

5.2 Why time-varying idiosyncratic risk matters

- If idiosyncratic risk rises in recessions, the insurance value of government debt rises in recessions.
- That countercyclical service flow pushes up bond prices exactly when aggregate marginal utility is high.
- This makes government debt a hedge and lowers the required cash-flow return.

Interpretation: The key mechanism is not just precautionary saving in levels. It is the covariance of safe-asset service flows with bad aggregate states.

5.3 What the quantitative exercise can and cannot prove

- The calibration shows that the mechanism is quantitatively plausible for US-style moments.
- It does not prove that all observed convenience yields are due to this mechanism.
- The model abstracts from explicit trading frictions and assumes liquid bond markets.
- The maximum sustainable deficit is sensitive to the target debt-output ratio and to the process for idiosyncratic risk.

- The government's unique ability to defend the safe-asset bubble is assumed rather than micro-founded fully within the model.

Interpretation: The paper is strongest as a theory of how safe-asset service flows should enter valuation and as a quantitative demonstration that the channel can be large.

6 Tables and figures

6.1 Figures 1 and 2: Safe-asset values and debt decomposition

Read: Figure 1:

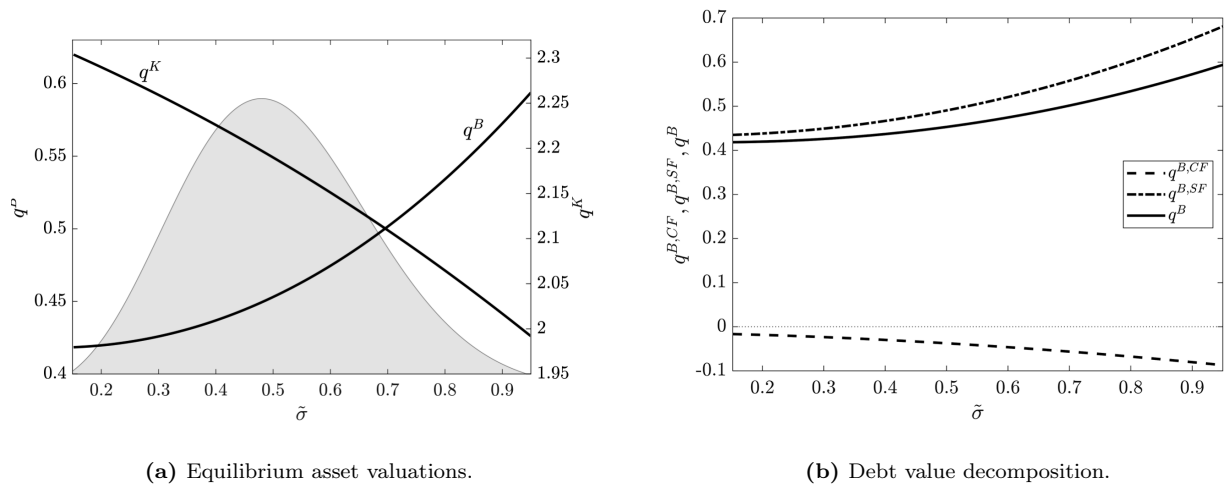
- The horizontal axis is idiosyncratic risk $\tilde{\sigma}$.
- The value of government debt q^B rises as idiosyncratic risk rises.
- The value of capital q^K falls as idiosyncratic risk rises.
- The shaded region is the stationary distribution of the risk state.

Read: Figure 2:

- The dashed cash-flow component $q^{B,CF}$ is negative and becomes more negative when idiosyncratic risk rises.
- The dash-dotted service-flow component $q^{B,SF}$ is positive and rises strongly with idiosyncratic risk.
- The total debt value q^B remains positive because the service-flow component dominates the negative cash-flow component.

Economic message: The two figures jointly show the paper's central force. Rising idiosyncratic risk raises households' demand for self-insurance, pushes up government bond value, and depresses capital value. Debt value is not mainly a claim on future primary surpluses; it is a claim that also produces insurance services.

Interpretation: A standard present-value test that focuses only on fiscal cash flows would infer overvaluation because $q^{B,CF}$ is negative. The dynamic-trading perspective resolves the puzzle by showing that households willingly hold government debt because the service-flow term is large and state-contingent.



Original figures directly cropped from the paper.

6.2 Table 1 and Table 2: calibration inputs and model fit

Read: Table 1:

- The key idiosyncratic-risk parameters are $\tilde{\sigma}^0 = 0.54$, mean reversion $\psi = 0.67$, and volatility $\sigma = 0.4$.
- The undiversifiable risk parameter is $\bar{x} = 0.3$.
- Risk aversion is set to $\gamma = 6$.
- Government expenditure and debt-policy parameters are chosen to support the quantitative debt and fiscal moments.

Read: Table 2:

- The model matches targeted macro moments closely: output volatility is 1.3% in both model and data; relative investment volatility is 3.35 versus 3.38.
- It also matches the average debt-output ratio, 0.74 in the model versus 0.71 in the data.
- The unlevered equity premium is 3.59% in the model versus 3.42% in the data, and the equity Sharpe ratio is 0.31 in both.
- Some untargted moments fit less well: the model overstates the volatility of the debt-output ratio and the average/volatility of the risk-free rate.

Economic message: The model is not tuned only to debt. It is disciplined by macro volatility, fiscal moments, debt valuation, and equity returns. The fit to the equity premium and Sharpe ratio matters because the same risk-pricing mechanism generates flight to safety into bonds.

Interpretation: The main weakness is the risk-free rate block. The model's risk-free rate is too high and too volatile. This does not eliminate the paper's safe-asset mechanism, but it signals that additional ingredients may be needed for a full quantitative asset-pricing model.

TABLE 1 PARAMETER CHOICES			TABLE 2 QUANTITATIVE MODEL FIT			
Parameter	Description	Value	Symbol	Description	Model	Data
$\tilde{\sigma}^0$	$\tilde{\sigma}_t^2$ stochastic steady state	.54	$\sigma(Y)$	Output volatility (%)	1.3	1.3
ψ	$\tilde{\sigma}_t^2$ mean reversion	.67	$\sigma(C)/\sigma(Y)$	Relative consumption volatility	.61	.64
σ	$\tilde{\sigma}_t^2$ volatility	.4	$\sigma(I)/\sigma(Y)$	Relative investment volatility	3.35	3.38
$\bar{x}$	Undiversifiable risk	.3	$\sigma(S/Y)$	Surplus volatility (%)	1.1	1.1
γ	Risk aversion	6	$E[C/Y]$	Average consumption-output ratio	.58	.56
ρ	Time preference	.138	$E[G/Y]$	Average government expenditures-output ratio	.22	.22
a^0	a_t stochastic steady state	.625	$E[S/Y]$	Average surplus-output ratio	-.0005	-.0005
g	Government expenditures	.138	$E[I/K]$	Average investment rate	.12	.12
$\tilde{\mu}^{S,0}$	$\tilde{\mu}_t^S$ stochastic steady state	.0026	$E[q^K K/Y]$	Average capital-output ratio	3.48	3.73
α^a	a_t slope	.072	$E[q^B K/Y]$	Average debt-output ratio	.74	.71
α^S	$\tilde{\mu}_t^S$ slope	.12	$E[d\tilde{r}^E - d\tilde{r}^S]$	Average (unlevered) equity premium (%)	3.59	3.42
ϕ	Capital adjustment cost	8.1	$E[d\tilde{r}^E - d\tilde{r}^S]/\sigma(d\tilde{r}^E - d\tilde{r}^S)$	Equity Sharpe ratio	.31	.31
ι^0	Capital adjustment intercept	-.022				
δ	Depreciation rate	.055				
			B. Untargeted Moments			
			$\rho(Y, C)$	Correlation of output and consumption	.98	.92
			$\rho(Y, I)$	Correlation of output and investment	.99	.94
			$\rho(Y, S/Y)$	Correlation of output and surpluses	.98	.60
			$\sigma(q^K K/Y)$	Volatility of debt-output ratio (%)	4.8	2.0
			$E[r^f]$	Average risk-free rate (%)	5.18	.69
			$\sigma(r^f)$	Volatility of risk-free rate (%)	5.47	2.43

(a) Parameter choices.

(b) Quantitative model fit.

Original tables directly cropped from the paper.

6.3 Figure 3: debt Laffer curve

Read:

- The horizontal axis varies the average interest-adjusted debt growth rate.
- The vertical axis is the sustainable primary deficit as a share of GDP.
- In the full dynamic model, the Laffer curve peaks at about 2.25% of GDP.
- The peak occurs when average debt growth is around 7.3%; beyond that point, additional debt issuance erodes the tax base.
- In the steady-state benchmark with constant idiosyncratic risk, the Laffer curve is tiny and peaks around 0.1% of GDP.

Economic message: The government can fund deficits by mining the safe-asset bubble, but only up to a point. A higher debt growth rate acts like a tax on bondholders' self-insurance. Excessive taxation destroys the value of the safe asset and shrinks the revenue base.

Interpretation: The difference between the solid and dotted curves is the paper's quantitative punchline. Countercyclical idiosyncratic risk and the resulting negative-beta property are not cosmetic; they are what make debt capacity large.

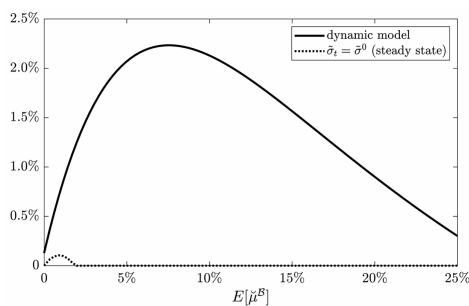


Figure 3: Debt Laffer curve, directly cropped from the paper.

7 Short summary

- The paper develops a theory of safe assets based on self-insurance through retrading.
- Government bonds are valuable because households face uninsurable idiosyncratic risk and can trade bonds after shocks.
- The value of a safe asset has two components: cash-flow value and service-flow value.
- The service-flow value rises when idiosyncratic risk rises.
- If idiosyncratic risk rises in recessions, government bonds can appreciate in recessions and have negative beta.
- The dynamic-trading valuation perspective resolves the debt valuation puzzle by including the service-flow component.
- Bubbles can be attached to safe assets and can reinforce safety by increasing the value of the service-flow component.
- Governments have a special role because they can defend a debt bubble through fiscal authority, regulation, and market-making capacity.

- Bubble mining can finance permanent deficits, but the government faces a debt Laffer curve.
- In the calibration, the dynamic model supports a maximum permanent deficit of about 2.25% of GDP, much larger than the constant-idiosyncratic-risk benchmark.

8 Conclusion

8.1 Core conclusion

Brunnermeier, Merkel, and Sannikov offer a theory in which safe assets are valuable because they allow households to self-insure through dynamic trading. This means that government debt is not valued only as a claim on primary surpluses. It also produces a service flow that becomes especially valuable when idiosyncratic risk is high. Once this service flow is recognized, government debt can have a negative beta, command a low cash-flow return, and appear to be worth more than the present value of fiscal cash flows.

8.2 Incremental contribution of the paper

The incremental contribution is the dynamic-trading perspective. The paper shows how to value the gains from equilibrium retrading in an incomplete-markets economy and applies that method to government debt. This differs from standard convenience-yield models because the service flow is not assumed directly in preferences. It is generated endogenously by households' desire to trade safe assets after private shocks.

8.3 Economic intuition

The mechanism is easiest to understand from the perspective of a household running a risky firm. The household cannot sell all idiosyncratic exposure because of skin-in-the-game constraints. Government bonds are therefore useful because they can be sold or bought after idiosyncratic shocks. When idiosyncratic risk rises, households value this trading option more. If that rise in idiosyncratic risk occurs in recessions, the bond becomes a hedge against bad aggregate states. This is why the safe asset can have a negative beta even when its direct cash flows are not especially attractive.

8.4 Government debt and the debt valuation puzzle

The paper's framework changes how one should read government debt valuation equations. A standard buy-and-hold formula says that debt should be backed by primary surpluses or by a terminal bubble. The dynamic-trading formula decomposes debt value into fiscal cash flows plus service flows. This matters because the paper's calibration finds that the fiscal cash-flow component can be negative while the service-flow component is positive and large. Debt is therefore not necessarily overvalued; rather, the standard valuation formula may be missing the safe-asset service flow.

8.5 Policy implication: safe-asset capacity is valuable but limited

The debt Laffer curve is central for policy. A government that issues a valued safe asset can finance permanent primary deficits by expanding debt. But the safe-asset franchise cannot be mined without limit. If debt issuance is too aggressive, investors require compensation for the dilution of their bond claims, the real value of debt falls, and revenue declines. The calibrated peak near 2.25% of GDP should therefore be interpreted as a debt-capacity bound, not as a recommendation to run deficits up to that point.

8.6 Financial-stability implication

The model assigns a special role to liquidity and market functioning, even though explicit trading frictions are abstracted away. The service flow exists only if households can actually trade government bonds when they need to. This makes central-bank market-making capacity and deep government bond markets part of safe-asset fundamentals. A loss of market liquidity can therefore threaten safe-asset status even if the government's fiscal position has not changed immediately.

8.7 Private safe assets versus public safe assets

Private assets can also provide insurance services to holders. The difference is that private issuers cannot credibly defend a bubble on their liabilities in the same way as the government. They face default, limited taxation power, and no comparable regulatory authority. The paper therefore provides a theory of why public debt has a distinctive exorbitant privilege even when private securities are also liquid.

8.8 Limits and open questions

The paper abstracts from explicit bid-ask spreads, dealer balance sheets, collateral constraints, and institutional demand from banks and foreign official investors. It also relies on the assumption that idiosyncratic risk is countercyclical and that government debt markets remain liquid. Future work can combine this dynamic-trading service-flow channel with liquidity microstructure, international reserve demand, banking regulation, and political constraints on fiscal policy.

8.9 Takeaway

Safe assets are safe not only because they promise stable cash flows. They are safe because they are useful to trade in bad states. Government debt can therefore be valuable, low yielding, and negative beta even when fiscal cash flows are weak. But this privilege is not unlimited: it depends on the credibility, liquidity, and risk-hedging capacity of the public debt market.